Morphology- and Vesicle-Aware Neural Unit Benchmark

Synthetic Identifiability and a Prospective Allen Cell Types Biological-Component Test

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Abstract

Common artificial-neuron abstractions compress a biological cell into a pointwise weighted sum and a scalar output. Biological neurons expose additional variables: nonlinear dendritic subunits, morphology-dependent attenuation, adaptation, action-potential waveform state, probabilistic vesicle release, and multiple release sites. The existence of those variables does not establish that every model should retain them. A richer unit earns scientific value only when the proposed variables are identifiable from appropriate observations and improve held-out prediction under controlled comparison.

This paper defines a morphology- and vesicle-aware candidate unit and reports two bounded tests. First, across 20 fixed synthetic seeds, models were trained on 24 simulated cells and evaluated on 12 disjoint held-out cells per seed. The full feature family achieved a median joint negative log likelihood of 0.206522 per time bin and beat six nested or destroyed-alignment comparators on all 20 seeds. Second, a prospective public-data component test used 510 mouse visual-cortex cells with paired Allen Cell Types morphology and electrophysiology summaries. A donor-separated hash split fixed 432 cells for development and 78 cells from 60 unseen donors for final testing before model performance was examined.

Adding five morphology summaries to transgenic-line, layer, dendrite-type, apical-status, and reporter metadata lowered the pooled donor-equal final standardized mean squared error by 0.01247 and improved 8 of 11 prespecified electrophysiology outcomes. The joined model also beat a destroyed morphology-to-cell alignment control by 0.01532 standardized mean squared error. However, the donor-cluster bootstrap 95 percent interval for the joined-versus-metadata difference was [-0.03618, 0.00930], so the strict biological-component confirmation gate did not pass.

The synthetic result is an identifiability sanity check, and the real-data result is a directional but statistically unresolved morphology signal. Neither validates waveform-conditioned vesicle release. The generator contains the same branch and release structure favored by the full synthetic family, the Allen table contains no joined presynaptic-calcium or vesicle-output measurement, and the comparator labels are bounded surrogates rather than complete implementations of published LIF, GLIF, Hodgkin-Huxley, or active-dendrite model classes. A formal marginalization result shows why release parameters cannot be identified from spike-only observations and specifies the measurements required for a decisive joined benchmark.

Keywords: computational neuroscience; dendritic computation; multivesicular release; synaptic transmission; neuron model; identifiability; held-out-cell benchmark; Self Aware Networks

Scope and evidence boundary

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1. Purpose and historical scope

The motivating question is not whether biological neurons are more complicated than artificial point units. They are. The useful question is whether a declared subset of that complexity improves prediction of measurements that simpler models cannot already explain.

Micah Blumberg's publicly archived February 2021 article argued that a one-site synaptic-unreliability picture can understate biological computation when multiple release sites, locally regulated release probability, and presynaptic waveform modulation are considered together [1]. A March 2021 discussion extended that inquiry from the synaptic bottleneck to the flow of information through dendrites, somata, axons, and downstream arrays [2]. The 2022 SAN notes then joined branch-sensitive integration, action-potential duration, calcium-channel opportunity, vesicle output, and morphology into a proposed richer neural unit [3-5].

The chronology must remain bounded. The 2021 article is a public conceptual and narrative bridge, not an implementation of the model in this paper. The cited 2022 repository versions contain more explicit mechanism claims, but they do not provide the present equations, generator, held-out-cell experiment, or result. A retrospective note that labels part of the idea as 2014 does not independently establish the full current operator in 2014; the immutable source used here is the 2022 repository record [4].

The scientific premise also has substantial prior art. Hodgkin and Huxley modeled voltage-dependent ionic conductances [6]. Nonlinear dendritic subunits can make a pyramidal neuron behave like a two-layer network [7]. GLIF models can reproduce held-out spike timing while remaining computationally compact [8], and systematic workflows can fit morphologically and biophysically detailed models to recorded cells [9]. Release probability varies between synapses and is dynamically regulated [10]. Multivesicular release is preparation- and state-dependent rather than a universal deterministic alphabet [11-13]. Presynaptic potassium-channel state and action-potential waveform can alter calcium entry and facilitation in specific preparations [14,15].

The contribution tested here is consequently narrow and staged. First: does a joined feature family containing branch-resolved nonlinear integration and waveform-conditioned vesicle output pass a synthetic identifiability gate that simpler nested families fail? Second: before vesicle-linked measurements exist in one public dataset, do coarse reconstructed-morphology summaries contribute any donor-held-out predictive information for cellular electrophysiology beyond declared metadata?

2. Candidate unit

Let a cell contain branches indexed by b, with input count $x_{b,t}$ in discrete time bin t. Each branch carries a leaky local state

$$h_{b,t} = \lambda_h h_{b,t-1} + x_{b,t}, \qquad 0 < \lambda_h < 1. \tag{1}$$

Let ℓ_b denote a normalized branch-length proxy. The branch attenuation and threshold are allowed to depend on that morphology. A nonlinear branch response is

$$z_{b,t} = \exp(-\kappa \ell_b) \left[\max(h_{b,t} - \theta_0 - \theta_1 \ell_b, 0)\right]^2 . \tag{2}$$

The cell also carries a spike-history adaptation state. If $s_t \in \{0, 1\}$ is the spike indicator, then

$$r_t = \lambda_r r_{t-1} + s_t. \tag{3}$$

The candidate spike probability combines the morphology-weighted branch state, a slow integrated drive g_t, and the prior adaptation state:

$$
\begin{aligned}
p_t^{\text{spike}} &= \sigma\left(\alpha_0 + \alpha_z \sum_b z_{b,t} + \alpha_g g_t - \alpha_r r_{t-1}\right), \\
s_t &\sim \text{Bernoulli}\left(p_t^{\text{spike}}\right),
\end{aligned}
\tag{4}
$$

where $\sigma(q) = (1 + e^{-q})^{-1}$. A waveform-duration proxy d_t is modeled only on the scale needed for this entry test:

$$
d_t = d_0 + d_r r_{t-1} + d_z \max_b z_{b,t} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_d^2). \tag{5}
$$

Conditional on a spike, the model assigns a release probability and a vesicle-count output

$$
\begin{aligned}
p_t^{\text{rel}} &= \sigma\left(\beta_0 + \beta_1 d_t + \beta_2 d_t^2 + \beta_3 \max_b z_{b,t} + \beta_4 m\right), \\
v_t \mid s_t = 1 &\sim \text{Binomial}\left(N, p_t^{\text{rel}}\right),
\end{aligned}
\tag{6}
$$

where m is a cell-level morphology index and N is the modeled number of available release opportunities. Equation (6) is not a claim that all synapses share one fixed N, that each vesicle produces an equal postsynaptic effect, or that waveform alone determines release. It is a minimal observation model for asking whether graded output can be recovered when vesicle counts are actually observed.

3. A spike-only non-identifiability result

Partition the parameters into spike parameters ϕ and release parameters ψ. For one time bin, the joined model factorizes as

$$
p(s_t, v_t \mid X_t; \phi, \psi) = p(s_t \mid X_t; \phi)\, [p(v_t \mid s_t = 1, X_t; \psi)]^{s_t}, \tag{7}
$$

with $v_t = 0$ by convention when no spike occurs. If the experiment observes only spikes, summing over the unobserved release count gives

$$
p(s_t \mid X_t; \phi, \psi) = \sum_{v_t=0}^{N} p(s_t, v_t \mid X_t; \phi, \psi) = p(s_t \mid X_t; \phi), \tag{8}
$$

because the conditional release distribution is normalized. The spike-only likelihood is therefore constant in ψ. No amount of spike-only fitting can identify the vesicle-count mechanism in Equation (6) without additional assumptions that couple release back into later observed spikes.

This is a measurement-design result, not merely a statistical inconvenience. A biological test of a vesicle-aware unit must measure a release-linked output, such as quantal glutamate transients or another validated proxy, jointly with presynaptic activity. A spike-only benchmark can evaluate branch integration and adaptation, but it cannot establish the release half of the model.

4. Synthetic held-out-cell benchmark

4.1 Generator and splits

The executable benchmark uses 20 fixed seeds. For each seed it generates 24 training cells and 12 disjoint test cells. Every cell contains four branches and 1,200 time bins. Branch lengths, attenuation, and thresholds vary by cell. Inputs combine sparse independent events, occasional branch bursts, and rarer multi-branch events. The target process implements Equations (1)-(6) with three release opportunities. Train and test cells are generated independently under the same declared family.

This construction tests whether the relevant variables are learnable across new cells when the target process contains them. It does not ask whether the generator is an adequate model of any biological cell type.

4.2 Comparator feature families

Each family is fitted with ridge-regularized logistic components using only its declared features.

Feature family	Spike-side information	Release-side information	Fitted parameters
LIF-like point	one fast summed-input state	constant release probability	4
GLIF-like adaptive	fast and slow summed states plus adaptation	constant release probability	6
Point-nonlinear waveform	summed states, scalar nonlinearity, change, adaptation	duration and squared duration	9
Dendritic-subunit	branch nonlinearities without morphology weights, slow state, adaptation	constant release probability	9
Linear state-space	linear branch states, summed states, adaptation	duration and branch peak	11
Morphology-vesicle full	morphology-weighted branch nonlinearities, slow state, adaptation	duration, nonlinear duration, branch peak, morphology	12
Full destroyed controls	full feature count, but aligned branch and release variables permuted	permuted release variables	12

The names *LIF-like* and *GLIF-like* describe the information available to these surrogate regressions. They are not faithful implementations of every dynamic, reset, fitting, or conductance mechanism in the corresponding published model classes. No full Hodgkin-Huxley, NEURON, Allen GLIF, or active-dendrite simulator is executed here.

4.3 Outcomes

For spike targets s_i and predicted probabilities $\hat{p}_i$, the per-bin Bernoulli negative log likelihood is

$$\mathcal{L}_{\text{spike}} = -\frac{1}{T}\sum_{i=1}^{T}\left[s_i \log \hat{p}_i + (1-s_i)\log(1-\hat{p}_i)\right]. \tag{9}$$

The release outcome uses the binomial negative log likelihood on spiking bins. To keep the two observation streams on a common time-bin scale, the primary joint score is

$$\mathcal{L}_{\mathrm{joint}} = \mathcal{L}_{\mathrm{spike}} + \frac{1}{T} \sum_{i:s_i=1} \left[-\log \mathrm{Binomial}\Big(v_i; N, \hat{p}_i^{\mathrm{rel}}\Big) \right]. \tag{10}$$

Lower values are better. Secondary outcomes are spike Brier score, release negative log likelihood per spike, and vesicle-count root mean square error. The script also verifies finite losses, nonempty test spike sets, disjoint train / test-cell construction, and prespecified direction checks.

5. Results

The full feature family achieved the lowest median primary loss. It beat every comparator on every one of the 20 held-out-cell seeds.

Feature family	Spike NLL / bin	Release NLL / spike	Vesicle RMSE	Joint NLL / bin	Other minus full
LIF-like point	0.169822	1.295731	0.922517	0.227671	0.022340
GLIF-like adaptive	0.167453	1.295731	0.922517	0.225073	0.019649
Point-nonlinear waveform	0.164960	1.124863	0.807474	0.214619	0.009343
Dendritic-subunit	0.159699	1.295731	0.922517	0.217319	0.011500
Linear state-space	0.167645	1.077869	0.782292	0.215721	0.010176
Morphology-vesicle full	**0.158355**	**1.077656**	**0.781143**	**0.206522**	-
Full destroyed controls	0.176796	1.285283	0.918660	0.234749	0.028010

Values are medians across the 20 seeds. *Other minus full* is the median seedwise difference in joint negative log likelihood; positive values favor the full family.

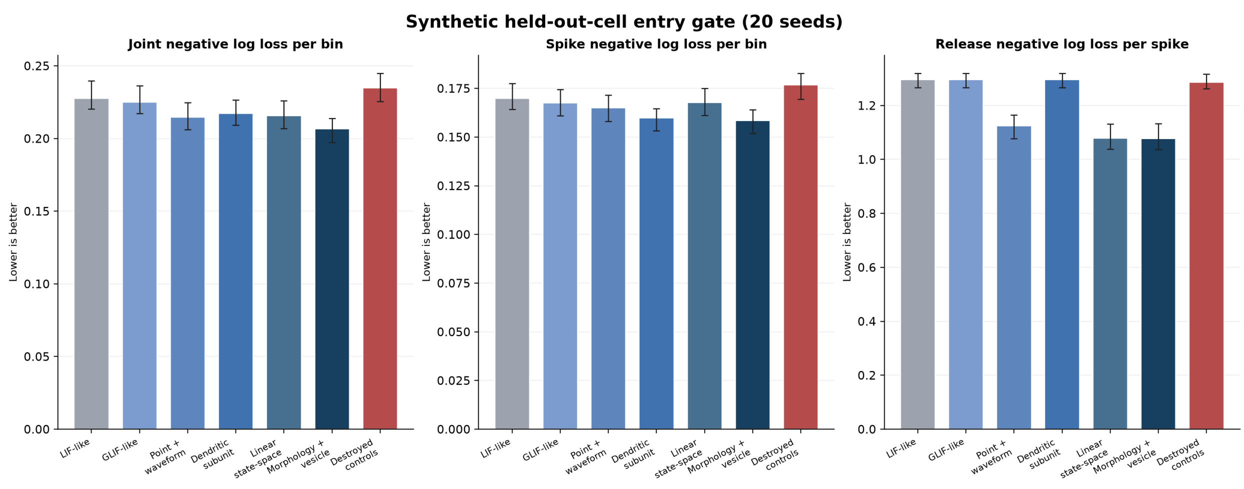

Figure 1: Three-panel summary of held-out spike loss, release loss, and joint loss across the seven benchmark feature families.

Figure 1. Synthetic held-out-cell entry benchmark. Points show the seedwise scores; bars mark medians. The full family benefits separately from branch-sensitive spike features and release-sensitive waveform features. Destroying their alignment removes the gain despite retaining the same feature count.

The ablation pattern is more informative than the rank alone. The dendritic family improves spike loss but cannot improve release loss because it has no release-side variables. The point-nonlinear waveform and linear state-space families improve release prediction, but lack the full morphology-weighted nonlinear branch description. The full family joins both information streams. When those features are independently permuted in the train and test sets, performance falls below even the simpler point baselines.

The result is not evidence that twelve parameters are universally preferable to fewer parameters. The test sets contain unseen cells, which limits ordinary within-sample overfitting, but the candidate still has greater representational capacity and is aligned with its own generator. Capacity-matched alternatives, broader hyperparameter searches, out-of-family generators, and real measurements are required before a model-class claim can be made.

6. Prospective Allen Cell Types biological-component test

6.1 Problem, public data, and frozen split

The synthetic gate shows that the proposed variables are recoverable when the generator contains them. The next question is whether morphology carries any predictive signal in real cells. The Allen Cell Types Database provides standardized intracellular current-clamp measurements and reconstructed neuronal morphology from mouse visual cortex, with computed electrophysiology and morphology summaries accessible through its public API [16,17].

Before examining model performance, the protocol froze every available mouse record with a morphology reconstruction and all five declared morphology fields: average contraction, average parent-daughter ratio, maximum Euclidean distance, number of bifurcations, and number of stems. This produced 510 cells from 372 donors. A SHA-256 rule assigned entire donors, rather than individual cells, to either development or final testing: 432 cells from 312 donors were used for development and 78 cells from 60 nonoverlapping donors were sealed for final testing.

All eleven electrophysiology summaries exposed by the API were prespecified as outcomes: adaptation, average firing rate, average interspike interval, f-I slope, fast-trough voltage, ramp peak time, input resistance, membrane time constant, threshold current, upstroke / downstroke ratio, and resting voltage. No outcome was selected or removed after its result was known.

Five fixed models were compared: a training-mean null, metadata-only ridge, morphology-only ridge, joined metadata-plus-morphology ridge, and a joined model whose morphology-to-cell alignment was destroyed by deterministic permutation. Metadata comprised transgenic line, cortical layer, dendrite type, apical tag, and reporter status. Ridge strength was selected separately for every outcome and feature family by donor-separated five-fold development error. The final outcomes remained unopened until the configuration, source hash, split manifest, code hash, and development lock were fixed.

For final cell i, outcome k, and model family f, standardized squared error was

$$e_{ik}^{(f)} = \left(\frac{y_{ik} - \widehat{y}_{ik}^{(f)}}{s_{k,\mathrm{dev}}} \right)^2 , \tag{11}$$

where $s_{k,\mathrm{dev}}$ is the development-set standard deviation for outcome k. To avoid allowing donors with more cells or more observed outcomes to dominate, the primary contrast first averaged within donor:

$$\Delta_d = \frac{1}{|I_d|} \sum_{(i,k)\in I_d} \left(e_{ik}^{(\mathrm{joined})} - e_{ik}^{(\mathrm{metadata})} \right) , \tag{12}$$

and then weighted the D final donors equally:

$$\overline{\Delta} = \frac{1}{D}\sum_{d=1}^{D}\Delta_d. \tag{13}$$

Negative values favor morphology. The strict gate required four conditions simultaneously: $\overline{\Delta} < 0$; a donor-cluster bootstrap 95 percent upper endpoint below zero; lower joined error on at least 6 of 11 outcomes; and lower joined error than the destroyed-alignment control.

6.2 Final-test results

Electrophysiology outcome	Final cells	Metadata MSE	Joined MSE	Destroyed MSE	Joined minus metadata
Adaptation	63	1.416227	1.399446	1.427063	-0.016781
Average firing rate	71	0.708635	0.715040	0.710179	+0.006405
Average interspike interval	71	0.369996	0.425936	0.372745	+0.055940
f-I curve slope	78	0.367980	0.375111	0.367561	+0.007131
Fast-trough voltage	78	0.637526	0.631620	0.601771	-0.005906
Ramp peak time	76	1.000916	0.964545	0.999275	-0.036371
Input resistance	78	0.858260	0.697914	0.866337	-0.160346
Membrane time constant	78	0.585506	0.559307	0.599881	-0.026199
Threshold current	78	0.941461	0.923540	0.951653	-0.017921
Upstroke / downstroke ratio	78	0.346015	0.331584	0.360312	-0.014431
Resting voltage	78	1.042815	1.028601	1.038209	-0.014214

The joined model had lower error than metadata alone on 8 of 11 outcomes. The donor-equal pooled difference was $\overline{\Delta} = -0.012468$, and the joined-minus-destroyed difference was -0.015323. Both point estimates favor correctly aligned morphology. The 10,000-resample donor-cluster bootstrap interval for $\overline{\Delta}$ was [-0.036183, 0.009300]. Because its upper endpoint crossed zero, three of the four prespecified conditions passed but the strict component gate did not.

Figure 2. Prospective Allen biological-component result. Negative bars favor the joined morphology-plus-metadata model. Directional gains occurred on eight outcomes and were largest for input resistance. The pooled estimate favored morphology, but the donor-bootstrap interval crossed zero; the result is promising rather than confirmatory.

6.3 Interpretation

This result advances the paper beyond a synthetic-only proposal without pretending that a failed uncertainty criterion is a positive confirmation. The sign pattern is consistent with morphology carrying modest, correctly aligned information about electrophysiological state across unseen donors. The especially large input-resistance improvement is biologically plausible, but it was not a separately prespecified primary endpoint and therefore remains one outcome within the complete eleven-outcome family.

The evidence is also narrower than the candidate unit. Five static reconstruction summaries are not branch-resolved voltage dynamics, and current-clamp feature prediction is not vesicle-output prediction. The public table cannot test Equation (6), because it does not join the same cells to bouton waveform, local calcium, and trial-resolved transmitter release. The strict gate's failure therefore directs the next experiment: increase donor-level power or use richer morphology while preserving a sealed evaluation, then acquire the release-linked observations required by the identifiability result.

7. What the evidence establishes

The synthetic benchmark establishes four bounded points.

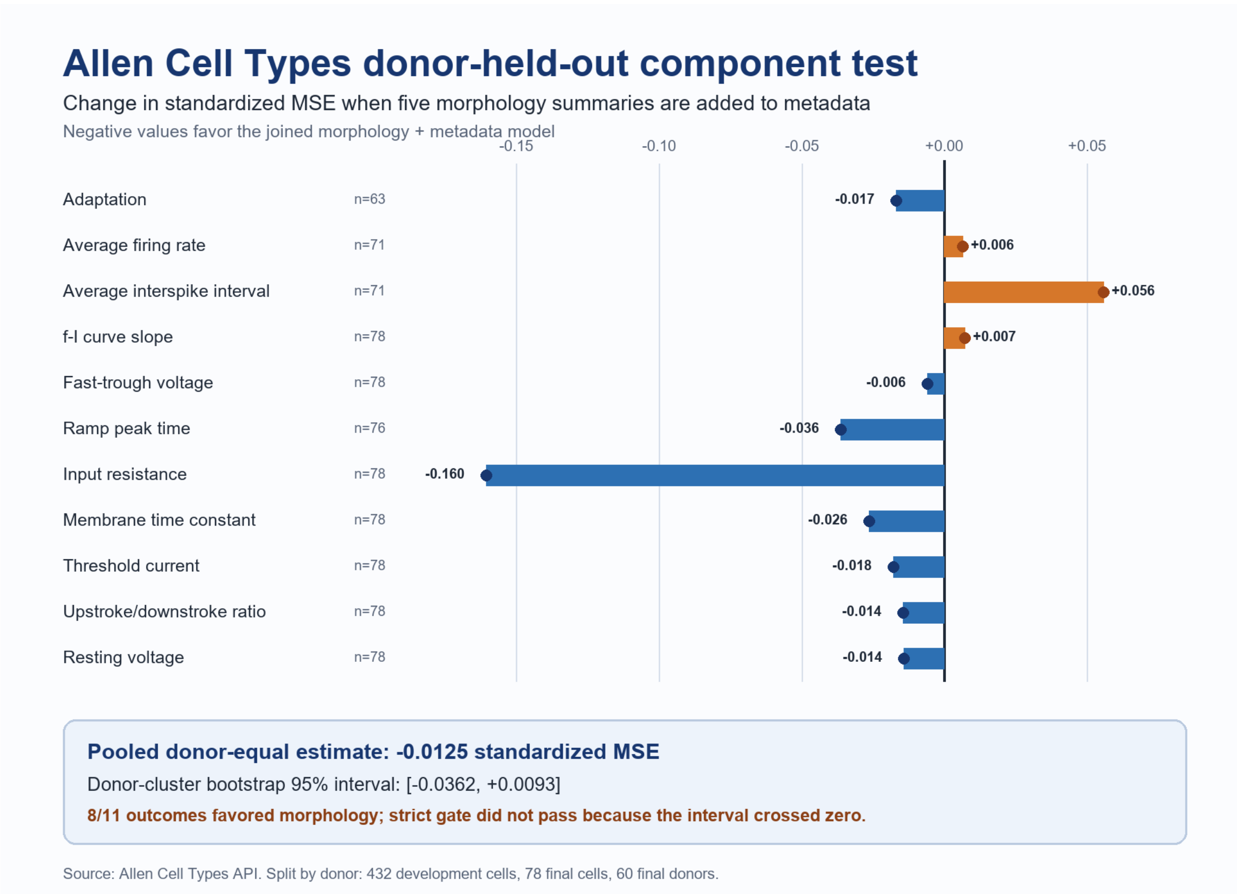

Figure 2: Final-test change in standardized mean squared error after adding five morphology summaries to the metadata model across eleven electrophysiology outcomes.

1. Branch-resolved and release-linked variables can be made jointly identifiable when the observation set contains branch inputs, spikes, waveform proxies, morphology, and vesicle counts.
2. A nested model that sees only spikes cannot identify the release mechanism, as Equation (8) predicts.
3. In this generator, dendritic features improve spike prediction and waveform / release features improve vesicle prediction; combining them improves the declared joint score.
4. The gain depends on correct alignment. Permutation controls reject the explanation that feature count alone produced the result.

The prospective Allen test adds three empirical findings.

1. Coarse morphology improved the final point estimate beyond declared metadata on 8 of 11 electrophysiology outcomes.
2. Correctly aligned morphology beat the destroyed-alignment control in the pooled donor-equal estimate.
3. The confidence interval included zero, so this protocol did not confirm a population-level morphology increment under its strict rule.

Together, the tests do not establish any of the following:

- statistically resolved or universal superiority on biological recordings;
- superiority to full LIF, GLIF, Hodgkin-Huxley, NEURON, active-dendrite, or state-space implementations;
- biological validation of the waveform-conditioned vesicle mechanism;
- a universal vesicle-count alphabet;
- deterministic synaptic transmission;
- a direct semantic code in action-potential duration or vesicle number;
- consciousness, neural rendering, NAPOT, or a complete SAN architecture;
- engineering advantage in an artificial-intelligence task.

Those claims require different data and different tests.

8. Required joined biological benchmark

The Allen test is the first biological component benchmark, but the decisive joined test must pre-register a cell type, synapse class, and preparation rather than pool incompatible synapses. Its minimum measurement set is:

- reconstructed branch morphology and branch-resolved stimulation or input inference;
- somatic voltage and spike timing;
- a presynaptic action-potential waveform measurement at the relevant bouton where technically possible;
- a calcium measurement or justified proxy at the same release site;
- trial-resolved transmitter output or vesicle-count proxy;
- repeated stimuli that estimate intrinsic variability and state dependence;
- held-out cells, held-out stimuli, and laboratory or session splits.

The comparison set should include an optimized point LIF model, the relevant Allen-style GLIF levels, a conductance-based Hodgkin-Huxley or comparable active model, a morphologically detailed compartmental model, an active-dendrite model, and a flexible state-space baseline. Each model should be evaluated both at matched parameter budget and at its best justified capacity. Runtime, memory, calibration, and uncertainty should be reported alongside predictive scores.

The decisive ablations are prospective. Shuffle morphology across cells, destroy branch identity while preserving total input, replace measured waveform with spike-only timing, remove calcium or release observations, and test whether morphology and release variables retain out-of-sample value. A richer unit fails if its gain disappears under capacity matching, does not transfer to new cells or stimuli, or depends on an observation unavailable at inference time. It also fails as a mechanistic account if different causal parameterizations predict the same observed outputs under all feasible interventions.

9. Relationship to SAN and artificial neurology

The 2021-to-2022 SAN source sequence poses a useful engineering challenge: artificial units may compress away branch, waveform, release, and local plasticity variables that matter to biological transformation [1-5]. This paper converts that challenge into a falsifiable entry gate. It does not assume that more biological detail is automatically better.

Within SAN, the proposed unit can serve as a molecular-to-network interface. Branch morphology describes a learned receptive structure; the spike process describes cell-level recruitment; the waveform proxy represents a bounded state variable; and vesicle release describes a probabilistic downstream output. Later phase, rendering, or semantic interpretations must be tested separately. Nothing in the present result proves that a vesicle count has a fixed meaning outside its receptor, pathway, and network context.

For artificial neurology, the practical research program is modular. The Allen result begins the first stage by testing morphology summaries, but does not close it. Next determine whether richer branch variables and measured release variables improve biological prediction. Then compress only the surviving variables into an efficient unit. Only after that should the unit be tested in machine-learning tasks against parameter-, compute-, and data-matched baselines. A negative result would be informative: it would show that the omitted biology is redundant for the target outcome, insufficiently observed, or expressed at a different scale.

10. Limitations

The synthetic benchmark is generated from the target architecture. Its positive result is therefore expected if the estimator and code are functioning. Its train / test split challenges generalization across random cells within one family, not distribution shift across species, cell types, preparations, or laboratories.

The Allen component benchmark uses real cells but only precomputed cross-sectional summaries. The 78 final cells come from unseen donors, yet all cells belong to one institutional pipeline and mouse visual-cortex collection. The analysis does not use raw sweeps, full SWC geometry, cell-type-balanced sampling, external laboratories, or interventions on morphology. Transgenic line and layer can absorb real biological structure; ridge regression can miss nonlinear morphology-function relations; and the bootstrap interval shows that donor-level uncertainty remains material. The result should therefore guide a larger or richer preregistered replication, not be promoted as proof of the full model.

The fitted synthetic models are logistic feature families rather than dynamical simulator replications. Parameter count is reported but not used as a penalty in the primary score. The generator uses four branches, three release opportunities, fixed bin width, simplified adaptation, a scalar morphology index, and a Gaussian waveform proxy. It omits conductance dynamics, cable properties, active channel distributions, receptor saturation, vesicle depletion and replenishment, facilitation and depression beyond the proxy state, neuromodulation, extracellular fields, and network recurrence.

The vesicle-count model assumes a binomial observation law. Real synaptic output can depart from that law because release sites and vesicles need not be independent or identical, measurements can saturate, and quantal size can vary. Multivesicular release also differs across synapse classes and extracellular conditions [11-13].

Finally, the benchmark tests predictive utility, not historical novelty. Dendritic computation, ion-channel dynamics, release probability, waveform-sensitive calcium entry, and multivesicular release have established prior literatures. The distinct claim advanced here is the value of testing their joined, bounded feature set under one explicit held-out-cell and destruction-control protocol.

11. Conclusion

A morphology- and vesicle-aware neural unit passes a synthetic identifiability gate when branch-resolved input and release-linked output are observed together. Across 20 fixed seeds, the joined feature family produced the lowest held-out-cell joint loss, and its advantage disappeared when the informative alignments were destroyed.

A prospective Allen component test then asked whether five real morphology summaries add donor-held-out information about cellular electrophysiology. The answer is encouraging but not decisive: 8 of 11 outcomes and both pooled point contrasts favored correctly aligned morphology, while the donor-bootstrap interval crossed zero. This is stronger than a synthetic-only argument and weaker than biological confirmation. It supports continued measurement and model development without validating the release half of the proposed unit.

The strongest conclusion remains methodological. A theory about graded synaptic output cannot be evaluated from spike trains alone, and a richer neuron should not be favored merely because biology is rich. Added state variables must predict measurements, survive capacity-matched alternatives and causal ablations, transfer across held-out cells and conditions, and earn their computational cost. The next gate is a higher-powered morphology replication followed by a joined bouton-waveform, calcium, and transmitter-output experiment.

Code and data availability

The release package contains the deterministic synthetic NumPy benchmark, complete per-seed results, figure-generation scripts, the frozen Allen component configuration, a target-free donor split manifest, the development lock, and complete final predictions. The downloaded Allen API response is bound by SHA-256 to its public source query. Normal and optimized Python replays produce the same recorded final-result hash. The Allen Cell Types data and documentation remain available from the Allen Institute [17].

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